Review: Key Concepts of Reinforcement Learning 📝

We have established a standardized formalism to describe any Reinforcement Learning problem, regardless of whether it is a simple Mars Rover or a complex autonomous helicopter.

The Core Components

Concept	Symbol	Definition
State	s	The current situation or position of the agent (e.g., Rover is in box 4).
Action	a	What the agent decides to do (e.g., Move Left).
Reward	$R(s)$	The immediate feedback received for being in a state (e.g., +100 for finding science, 0 for empty rock).
Discount Factor	γ	A number between 0 and 1 (e.g., 0.99). It determines how much we value future rewards compared to immediate rewards.
Return		The total sum of discounted future rewards. The goal is to maximize this.
Policy	$\pi(s)$	The function (or "brain") that maps a state to an action. It tells the agent *what to do* in any given situation.

The Markov Decision Process (MDP) 🧠

This entire formalism—States, Actions, Rewards, and the transition logic—is collectively called a **Markov Decision Process (MDP)**.

The "Markov" Property

The term "Markov" implies a specific property about the system:

- **The future depends *only* on the current state.**
- It does **not** depend on the history of how you got there.
- *Example:* In chess, to decide the next best move, you only need to look at the current board. It does not matter *how* the pieces arrived at those positions; the history is irrelevant to the decision-making process.

The MDP Cycle

The MDP represents a continuous loop of interaction:

1. **Agent** observes the current state (s).
2. **Agent** uses its Policy (π) to choose an Action (a).
3. **Environment** reacts to the action.
4. **Environment** returns a new State (s') and a Reward (R) to the Agent.
5. Repeat.

Applications of the Formalism

The exact same mathematical framework applies to vastly different real-world problems.

Application	State (s)	Action (a)	Reward (R)
Mars Rover	Position (1-6)	Left, Right	+100 (Science), 0 (Nothing)
Helicopter	Position, Orientation, Speed, Rotation	Stick movements	+1 (Stable), -1000 (Crash)
Chess	Position of all pieces on the board	Legal moves	+1 (Win), -1 (Loss), 0 (Tie)

Note on Discount Factors:

- **Rover Example:** We used $\gamma = 0.5$ (very impatient) for illustration.
- **Real World:** We usually use $\gamma \approx 0.99$ or 0.999 (patient), as we care about the long-term outcome (winning the game or keeping the helicopter flying).